

Exercise 5

Fix $n \geq 1$, $\tau \in (0, 1)$. For any vector $v \in \mathbb{R}^n$, define $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$, where $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$ are the order statistics of v . Under what condition is it true that, for all $v \in \mathbb{R}^n$, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$?

Solution The order statistics of $-v$ are given by $(-v)_{(i)} = -v_{(n-i+1)}$ for $i = 1, \dots, n$. Therefore, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$ holds for a vector $v \in \mathbb{R}^n$ if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n-\lceil \tau n \rceil+1)}) = v_{(n-\lceil \tau n \rceil+1)}.$$

To make this work for all $v \in \mathbb{R}^n$, the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when $\tau = 1/2$ and n is odd. ///